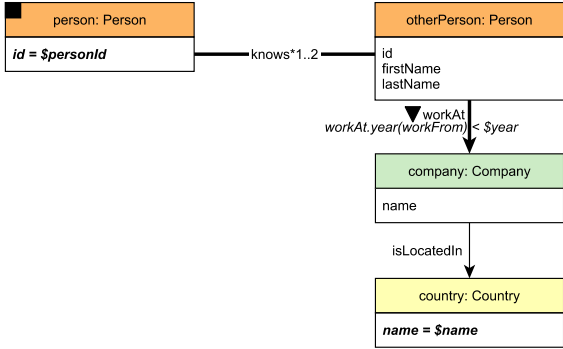


Interactive / complex / 11

IC 1	query	Interactive / complex / 11			
IC 2	title	Job referral			
IC 3	pattern	 <pre> graph TD P1[person: Person id = \$personId] -- knows*1..2 --> P2[otherPerson: Person id firstName lastName] P2 -- "workAt workAt.year(workFrom) < \$year" --> C[company: Company name] C -- isLocatedIn --> CO[country: Country name = \$name] </pre>			
IC 4					
IC 5					
IC 6					
IC 7					
IC 8					
IC 9					
IC 10					
IC 11					
IC 12					
IC 13					
IC 14v1	description	Given a start Person with ID \$personId, find that Person's friends and friends of friends (excluding start Person) who started working in some Company in a given Country with name \$countryName, before a given date (\$workFromYear).			
IC 14v2	params	1	\$personId	ID	
		2	\$countryName	String	
		3	\$workFromYear	32-bit Integer	
	result	1	otherPerson.id	ID	R
		2	otherPerson.firstName	String	R
		3	otherPerson.lastName	String	R
		4	company.name	String	R
		5	workAt.workFrom	32-bit Integer	R
	sort	1	workAt.workFrom	↑	
		2	otherPerson.id	↑	
		3	company.name	↓	
	limit	10			
	CPs	1.3, 2.3, 2.4, 3.3, 4.2			
	relevance	This query looks for paths of length two or three, starting from a Person, moving to friends or friends of friends, and ending at a Company. In this query, there are selective joins and a top-k order by that can be exploited for optimizations.			